



2012 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 2

Monday 20th February 2012

General Instructions

- Writing time — 2 Hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Collection

Section I Questions 1–10

- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.

Section II Questions 11–14

- Start each of these questions in a new booklet.
- Write your candidate number clearly on each booklet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Candidature — 88 boys

Examiner

KWM

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

Question One

Which of the following represents the complex number $1 - i$ expressed in modulus–argument form?

1

(A) $\sqrt{2} \operatorname{cis} \frac{3\pi}{4}$

(B) $\sqrt{2} \operatorname{cis} \frac{\pi}{4}$

(C) $\sqrt{2} \operatorname{cis}(-\frac{\pi}{4})$

(D) $2 \operatorname{cis} \frac{\pi}{4}$

Question Two

Which of the following is true for all complex numbers z ?

1

(A) $\operatorname{Re}(z) = z + \bar{z}$

(B) $\operatorname{Re}(z) = \frac{z - \bar{z}}{2}$

(C) $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$

(D) $\operatorname{Re}(z) = \frac{z\bar{z}}{2}$

Question Three

The primitive of $\frac{1}{\sqrt{4 - 9x^2}}$ is:

1

(A) $\frac{1}{6} \sin^{-1} \frac{3x}{2} + C$

(B) $\frac{1}{3} \sin^{-1} \frac{3x}{2} + C$

(C) $\frac{2}{3} \sin^{-1} \frac{3x}{2} + C$

(D) $\frac{1}{3} \sin^{-1} \frac{2x}{3} + C$

Exam continues next page ...

Question Four

The solutions to the quadratic equation $z^2 - 2z + 2 = 0$ are:

1

- (A) $z = 1 + i$ or $1 - i$
- (B) $z = -1 + i$ or $-1 - i$
- (C) $z = \pm(1 + i)$
- (D) $z = 1 \pm 2i$

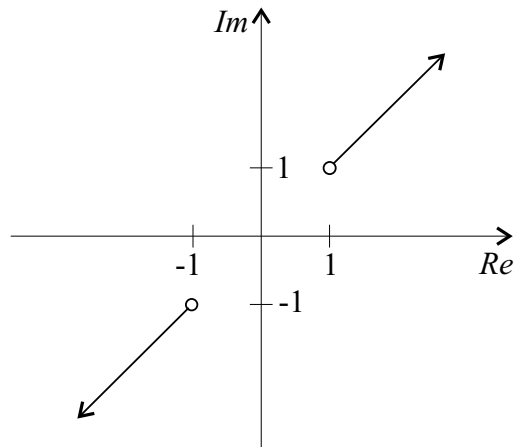
Question Five

The expression $i^{19} + i^{20} + i^{21} + i^{22}$ is equal to:

1

- (A) i
- (B) 0
- (C) $-i$
- (D) -1

Question Six



1

Which of the following defines the locus of the complex number z sketched in the diagram above?

- (A) $\arg\left(\frac{z - 1 + i}{z - 1 - i}\right) = \pi$
- (B) $\arg(z - 1 + i) = \arg(z - 1 - i)$
- (C) $\arg\left(\frac{z + 1 + i}{z - 1 - i}\right) = \frac{\pi}{2}$
- (D) $\arg(z + 1 + i) = \arg(z - 1 - i)$

Question Seven

What is the minimum value of $2 \sin x - 3 \cos x$?

1

- (A) -1
- (B) $-\sqrt{13}$
- (C) -5
- (D) $-\sqrt{2} - \sqrt{3}$

Question Eight

What is the primitive of $\frac{\cos \sqrt{x}}{\sqrt{x}}$?

1

- (A) $\sin \sqrt{x} + C$
- (B) $-\sin \sqrt{x} + C$
- (C) $\frac{1}{2} \cos^2 \sqrt{x} + C$
- (D) $2 \sin \sqrt{x} + C$

Question Nine

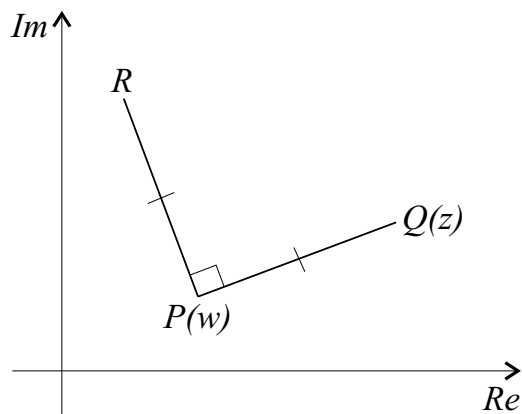
What is the range of $g(x) = \tan(\frac{1}{2} \cos^{-1} x)$?

1

- (A) $y \leq 0$
- (B) $y \in \mathbf{R}$
- (C) $0 \leq y < 1$
- (D) $y \geq 0$

Question Ten

1



In the diagram, $\angle QPR = 90^\circ$ and $PQ = PR$. The points P and Q represent the complex numbers w and z respectively. Which complex number represents the point R in the complex plane?

- (A) $i(w - z)$
- (B) $w + i(z - w)$
- (C) $w + i(w - z)$
- (D) $i(z - w)$

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

Question Eleven (15 marks) Use a separate writing booklet.	Marks
(a) Let $u = 2 + i$ and $v = -1 - i$.	
(i) Find $\text{Im}(uv)$.	1
(ii) Find $\frac{u}{i}$.	1
(iii) Evaluate $ u - v ^2$.	1
(iv) Express $\frac{u}{v}$ in the form $a + ib$, where a and b are real numbers.	2
(b) (i) Express the complex numbers $z_1 = 1 + i$ and $z_2 = \sqrt{3} - i$ in modulus-argument form.	1
(ii) Hence, or otherwise, find $\arg\left(\frac{z_1^2}{z_2}\right)$.	1
(c) (i) Find the two square roots of $16 + 30i$.	2
(ii) Hence solve $z^2 + (1 + i)z - (4 + 7i) = 0$.	2
(d) (i) Sketch the region in the complex plane which simultaneously satisfies $-\frac{\pi}{4} \leq \arg(z + i) \leq \frac{\pi}{4}$ and $ z - 2 \leq 1$.	2
(ii) Find the maximum value of $ z $.	1
(iii) Find the minimum value of $\arg(z)$, given $-\pi < \arg(z) \leq \pi$.	1

Question Twelve (15 marks) Use a separate writing booklet.**Marks**

(a) Find the following indefinite integrals.

(i) $\int \frac{1}{\sqrt{9x^2 + 2}} dx$ (Use the substitution $u = 3x$.) 2

(ii) $\int \frac{1}{x + \sqrt{x}} dx$ (Use the substitution $u = \sqrt{x}$.) 2

(b) Evaluate:

(i) $\int_0^1 \frac{1+x}{\sqrt{1-x^2}} dx$ 2

(ii) $\int_0^1 \frac{x^2 - 1}{x^2 + 1} dx$ 2

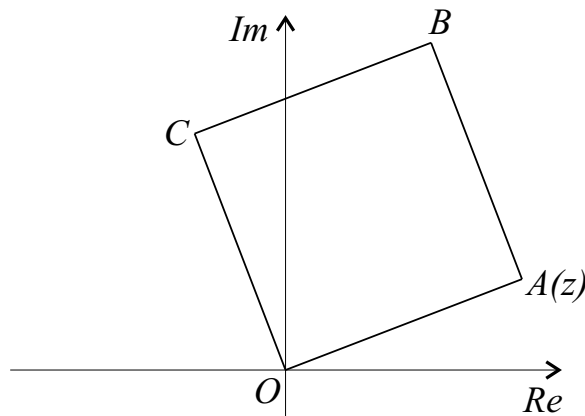
(c) Use integration by parts to find $\int x \ln x dx$. 2

(d) (i) Express $\frac{4x+2}{(x+1)(x^2+1)}$ in the form $\frac{a}{x+1} + \frac{bx+c}{x^2+1}$ 2
for real constants a , b and c .

(ii) Hence, or otherwise, evaluate $\int_0^{\sqrt{3}} \frac{4x+2}{(x+1)(x^2+1)} dx$. 3

Question Thirteen (15 marks) Use a separate writing booklet.**Marks**(a) Find the equation of the tangent to the curve $x^3 + y^3 - 5y - 3 = 0$ at the point $(1, -2)$. 3

(b)



In the diagram $OABC$ is a square where O is the origin. The point A represents the complex number z .

(i) Find the complex number represented by B in terms of z . 1(ii) The square is rotated about O anti-clockwise through 45° to $OA'B'C'$. 2Show that the point C' represents the complex number $\frac{\sqrt{2}}{2}(-1 + i)z$.

(c) The origin O and the points P , Q and R , representing the complex numbers z , $z + \frac{1}{z}$ and $\frac{1}{z}$ respectively, are joined to form a quadrilateral. Suppose that $0 < \arg(z) < \frac{\pi}{2}$ and $|z| \geq 1$.

(i) Sketch the quadrilateral $OPQR$ on the complex plane. 2(ii) Find the complex number z for $OPQR$ to be a square. 1(d) (i) Show that $(k + 1)^2(k + 4) = k^3 + 6k^2 + 9k + 4$. 1(ii) Use mathematical induction to prove that for all integers $n \geq 1$, 4

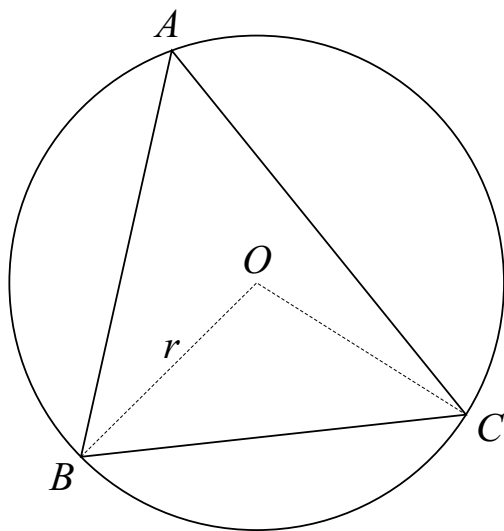
$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}.$$

(iii) Hence find 1

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{r(r+1)(r+2)}.$$

Question Fourteen (15 marks) Use a separate writing booklet.**Marks**(a) (i) Find a general solution to the equation $\cos 3x = \sin 2x$. 3(ii) Hence, or otherwise, find the smallest positive solution of the equation $\cos 3x - \sin 2x = 0$. 1

(b)

The circle drawn through the vertices of triangle ABC has centre O and radius r .(i) Show that $BC = 2r \sin A$. 2(ii) Use the fact that the sum of the areas of triangles OBC , OCA and OAB is equal to the area of triangle ABC , to show that 2

$$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C.$$

(c) (i) If $z = \cos \theta + i \sin \theta$, show that $1 + z = 2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$. 1(ii) Suppose that z_1 and z_2 are complex numbers such that $|z_1| = |z_2| = 1$. If z_1 and z_2 have arguments α and β respectively, where $-\pi < \alpha < \pi$ and $-\pi < \beta < \pi$, show that $\frac{z_1 + z_1 z_2}{z_1 + 1}$ has modulus $\frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}}$ and argument $\frac{\alpha + \beta}{2}$. 2(iii) Given that $|z_1| = |z_2| = 1$ and $\frac{z_1 + z_1 z_2}{z_1 + 1} = 2i$, find z_1 and z_2 in the form $x + iy$, where x and y are real. 4

 End of Section II

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SYDNEY GRAMMAR SCHOOL



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

QUESTION 1

$$\left. \begin{aligned} r \cos \theta &= 1 \\ r \sin \theta &= -1 \end{aligned} \right\} \begin{aligned} r &= \sqrt{2} \\ \theta &= -\pi/4 \end{aligned}$$

$$\sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$

C

QUESTION 2

let $z = a + ib$
 $\bar{z} = a - ib$
 $\frac{z + \bar{z}}{2} = \frac{a + ib + a - ib}{2}$
 $= a$
 $= \operatorname{Re}(z)$

C

QUESTION 3

$$\int \frac{1}{\sqrt{4-9x^2}} dx = \frac{1}{3} \sin^{-1} \frac{3x}{2} + c$$

B

QUESTION 4

$$\begin{aligned} z^2 - 2z + 2 &= 0 \\ (z-1)^2 + 1 &= 0 \\ (z-1)^2 &= -1 \\ z-1 &= \pm i \\ z &= 1+i \quad \text{or} \quad z = 1-i \end{aligned}$$

A

QUESTION 5

$$\begin{aligned} i^{19} + i^{20} + i^{21} + i^{22} \\ &= i^{19}(1 + i + i^2 + i^3) \\ &= i^{19}(1 + i - 1 - i) \\ &= 0 \end{aligned}$$

B

QUESTION 6

$$\begin{aligned} \arg(z - (1+i)) &= \arg(z - (-1-i)) \\ \arg(z - 1 - i) &= \arg(z + 1 + i) \\ \arg(z + 1 + i) &= \arg(z - 1 - i) \end{aligned}$$

D

QUESTION 7

let $2 \sin x - 3 \cos x = R \sin(x - \alpha)$
 $\left. \begin{aligned} R \cos \alpha &= 2 \\ R \sin \alpha &= 3 \end{aligned} \right\} R = \sqrt{13}$
 minimum value is $-\sqrt{13}$

B

QUESTION 8

let $u = \sqrt{x}$
 $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$
 $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = 2 \int \cos u du$
 $= 2 \sin u + c$
 $= 2 \sin \sqrt{x} + c$

D

QUESTION 9

$$\begin{aligned} \cos^{-1} x \text{ range } [0, \pi] \\ \frac{1}{2} \cos^{-1} x \text{ range } [0, \frac{\pi}{2}] \\ \tan\left(\frac{1}{2} \cos^{-1} x\right) \text{ (graph)} \\ \text{range } y \geq 0 \end{aligned}$$

D

QUESTION 10

$$\begin{aligned} PQ &= z - w \\ PR &= i(z - w) \\ OR &= OP + PR \\ OR &= w + i(z - w) \end{aligned}$$

B

QUESTION 11

$$(a) \quad u = 2+i \quad v = -1-i$$

$$(i) \quad uv = (2+i)(-1-i) \\ = -2 - 2i - i + 1 \\ = -1 - 3i$$

$$\operatorname{Im}(-1-3i) = -3 \checkmark$$

$$(ii) \quad \frac{u}{i} = \frac{2+i}{i} \times \frac{i}{i} \\ = \frac{2i-1}{-1}$$

$$= 1-2i \checkmark$$

$$(iii) \quad |u-v|^2 = |3+2i|^2 \\ = 9+4 \\ = 13 \checkmark$$

$$(iv) \quad \frac{u}{v} = \frac{2+i}{-1-i} \times \frac{-1+i}{-1+i} \\ = \frac{-2+2i-i-1}{1+1} \\ = \frac{i-3}{2}$$

$$= -\frac{3}{2} + \frac{1}{2}i \checkmark$$

$$(b) \quad (i) \quad z_1 = 1+i \quad z_2 = \sqrt{3}-i \\ z_1 = \sqrt{2} \operatorname{cis} \frac{\pi}{4} \quad z_2 = 2 \operatorname{cis} \left(-\frac{\pi}{6}\right)$$

$$(ii) \quad \arg\left(\frac{z_1}{z_2}\right) =$$

$$= 2 \arg(z_1) - \arg(z_2)$$

$$= 2 \times \frac{\pi}{4} - -\frac{\pi}{6}$$

$$= \frac{\pi}{2} + \frac{\pi}{6}$$

(15)

$$= \frac{2\pi}{3} \checkmark$$

$$(c) \quad (a+bi)^2 = 16+30i$$

$$(i) \quad a^2 - b^2 + 2abi = 16+30i$$

$$\left. \begin{aligned} a^2 - b^2 &= 16 \\ 2ab &= 30 \end{aligned} \right\} \checkmark$$

$$a = 5 \text{ and } b = 3 \text{ or}$$

$$a = -5 \text{ and } b = -3$$

the square roots of $16+30i$ are $\pm(5+3i) \checkmark$

$$(ii) \quad z^2 + (1+i)z - (4+7i) = 0 \\ z = \frac{-(1+i) \pm \sqrt{(1+i)^2 + 4(4+7i)}}{2}$$

$$z = \frac{-(1+i) \pm \sqrt{16+30i}}{2}$$

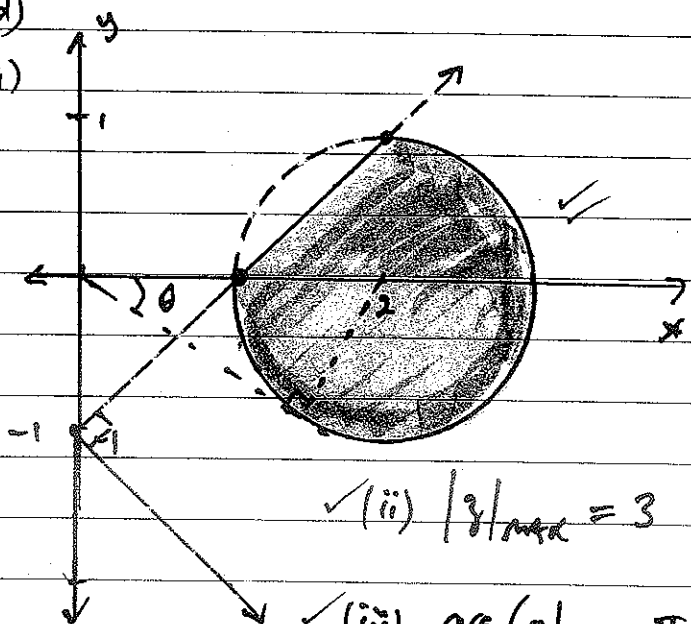
$$z = \frac{-1-i \pm (5+3i)}{2}$$

$$z = \frac{4+2i}{2} \text{ or } z = \frac{-6-4i}{2}$$

$$z = 2+i \text{ or } z = -3-2i$$

(d)

(i)



$$\checkmark (ii) \quad |z|_{\max} = 3$$

$$\checkmark (iii) \quad \arg(z) = -\frac{\pi}{6}$$

QUESTION 12

(a) (i)

$$\int \frac{1}{\sqrt{9x^2+2}} dx$$

$$\text{let } u = 3x \\ du = 3 dx$$

$$= \frac{1}{3} \int \frac{1}{\sqrt{u^2+2}} du \checkmark$$

$$= \frac{1}{3} \ln(u + \sqrt{u^2+2}) + c$$

$$= \frac{1}{3} \ln(3x + \sqrt{9x^2+2}) + c \checkmark$$

(ii)

$$\int \frac{1}{x+\sqrt{x}} dx$$

$$\text{let } u = \sqrt{x} \\ du = \frac{1}{2\sqrt{x}} dx$$

$$2u du = dx$$

$$= \int \frac{1}{u^2+u} \times 2u du \checkmark$$

$$= 2 \int \frac{1}{u+1} du$$

$$= 2 \ln(u+1) + c$$

$$= 2 \ln(\sqrt{x}+1) + c \checkmark$$

(b) (i)

$$\int_0^1 \frac{1+x}{\sqrt{1-x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} dx$$

$$= \left[\sin^{-1}x - \sqrt{1-x^2} \right]_0^1$$

$$= \frac{\pi}{2} + 1 \checkmark$$

(ii)

$$\int_0^1 \frac{x^2-1}{x^2+1} dx$$

$$= \int_0^1 \frac{x^2+1-2}{x^2+1} dx$$

$$= \int_0^1 1 - \frac{2}{1+x^2} dx \checkmark$$

$$= \left[x - 2 \tan^{-1}x \right]_0^1$$

$$= \left(1 - \frac{\pi}{2} \right) - 0$$

$$= \frac{1}{2} (2 - \pi) \checkmark$$

(c)

$$\int x \ln x dx$$

$$\text{let } u = \ln x \quad v' = x \\ u' = \frac{1}{x} \quad v = \frac{x^2}{2}$$

--- (c)

$$\begin{aligned}
 \int x \ln x \, dx &= \frac{x^2}{2} \ln x - \int \frac{1}{x} \times \frac{x^2}{2} \, dx \\
 &= \frac{x^2}{2} \ln x - \int \frac{x}{2} \, dx \\
 &= \frac{1}{2} x^2 \ln x - \frac{x^2}{4} + c
 \end{aligned}$$

$$= \left[\frac{1}{2} \ln(x^2+1) + 3 \tan^{-1} x - \ln(x+1) \right]$$

$$= \frac{1}{2} \ln 4 + \pi - \ln(\sqrt{3}+1)$$

$$= \ln \left(\frac{2}{\sqrt{3}+1} \right) + \pi$$

$$= \ln \left(\frac{2 \times \sqrt{3}-1}{\sqrt{3}+1 \times \sqrt{3}-1} \right) + \pi$$

$$= \ln \left(\frac{2\sqrt{3}-2}{3-1} \right) + \pi$$

$$= \ln(\sqrt{3}-1) + \pi$$

(d)

(i) let

$$\frac{4x+2}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$

$$4x+2 \equiv a(x^2+1) + (x+1)(bx+c)$$

put $x = -1$:

$$-2 = 2a, \quad \underline{a = -1}$$

(15)

$$4x+2 = (a+b)x^2 + (b+c)x + (a+c)$$

equating powers of x :

$$\begin{aligned}
 a+b &= 0 \\
 b+c &= 4 \\
 a+c &= 2
 \end{aligned}
 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \underline{b=1} \\ \underline{c=3} \end{array} \quad \checkmark$$

(ii)

$$\int_0^{\sqrt{3}} \frac{4x+2}{(x+1)(x^2+1)} \, dx$$

$$= \int_0^{\sqrt{3}} \frac{-1}{x+1} + \frac{x+3}{x^2+1} \, dx$$

$$= \int_0^{\sqrt{3}} \frac{x}{x^2+1} + \frac{3}{x^2+1} - \frac{1}{x+1} \, dx$$

QUESTION 13

(a)

$$x^3 + y^3 - 5y - 3 = 0$$

$$3x^2 + 3y^2 \frac{dy}{dx} - 5 \frac{dy}{dx} = 0 \quad \checkmark$$

$$\frac{dy}{dx} (3y^2 - 5) = -3x^2$$

$$\frac{dy}{dx} = \frac{3x^2}{5 - 3y^2} \quad \checkmark$$

at $(1, -2)$

$$\text{gradient} = \frac{3}{-7}$$

equatⁿ of the tangent:

$$y - y_1 = m(x - x_1)$$

$$y + 2 = \frac{-3}{7}(x - 1) \quad \checkmark$$

$$7y + 14 = -3x + 3$$

$$3x + 7y + 11 = 0$$

(b)

(i) B:

$$OC = AB = iz$$

$$OB = OA + AB$$

$$= z + iz \quad \checkmark$$

(ii)

$$OC' = OC \times \text{cis } \frac{\pi}{4}$$

$$= iz \text{ cis } \frac{\pi}{4} \quad \checkmark$$

$$= iz \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$$

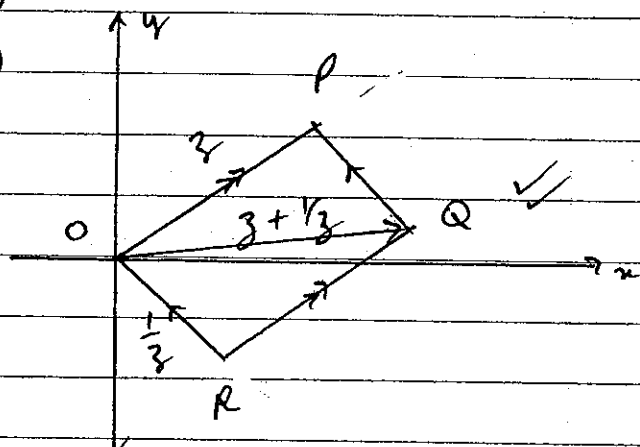
$$= iz \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i}{2} \right)$$

$$= \frac{\sqrt{2}}{2} iz - \frac{\sqrt{2}}{2} z \quad \checkmark$$

$$= \frac{\sqrt{2}}{2} (-1 + i) z$$

(c)

(i)

note: $\arg(z) = -\arg(1/z)$ (ii) $|z| = 1$ and $\arg(z) = \frac{\pi}{4}$ Hence $z = \text{cis } \frac{\pi}{4}$ for
OPQR to be a square.

(d)

(i)

$$\text{LHS} = (k+1)^2 (k+4)$$

$$= (k^2 + 2k + 1)(k+4)$$

$$= k^3 + 4k^2 + k^2 + 8k$$

$$+ k + 4$$

$$= k^3 + 6k^2 + 9k + 4$$

$$= \text{RHS.} \quad \checkmark$$

(ii)

A/ Show the statement
is true for $n=1$.put $n=1$:

$$\text{LHS} = \frac{1}{1 \times 2 \times 3}$$

$$= \frac{1}{6} \quad \checkmark$$

Q13...

6.

$$\begin{aligned}
 \text{RHS} &= \frac{1(1+3)}{4(1+1)(1+2)} \\
 &= \frac{4}{4 \times 2 \times 3} \\
 &= \frac{1}{6}
 \end{aligned}$$

Hence the statement is true for $n=1$.

B₁ Assume true for $n=k$. (k an integer.)

i.e.

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \dots + \frac{1}{k(k+1)(k+2)} = \frac{k(k+3)}{4(k+1)(k+2)} \quad (*)$$

To prove for $n=k+1$.

To prove:

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \dots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} = \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

Using the assumption (*)

$$\begin{aligned}
 \text{LHS} &= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \quad \checkmark \\
 &= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)} \\
 &= \frac{k(k^2 + 6k + 9) + 4}{4(k+1)(k+2)(k+3)} \\
 &= \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)} \quad (\text{from part (i)}) \\
 &= \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)} \\
 &= \frac{(k+1)(k+4)}{4(k+2)(k+3)} \\
 &= \text{RHS.}
 \end{aligned}$$

Q 13 --- (a)(ii)

7.

C/ It follows from parts A and B by mathematical induction that the statement is true for all integers $n \geq 1$. ✓

$$(ii) \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{r(r+1)(r+2)}$$

$$= \lim_{n \rightarrow \infty} \frac{n(n+3)}{4(n+1)(n+2)}$$

Divide by n^2 :

$$= \lim_{n \rightarrow \infty} \frac{1(1 + \frac{3}{n})}{4(1 + \frac{1}{n})(1 + \frac{2}{n})} \quad \checkmark$$

(as $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$)

$$= \frac{1}{4} //$$

(15)

QUESTION 14

(a)

(i) $\cos 3x = \sin 2x$

$$\cos 3x = \cos\left(\frac{\pi}{2} - 2x\right) \checkmark$$

$$3x = 2\pi n + \left(\frac{\pi}{2} - 2x\right)$$

$$5x = 2\pi n + \frac{\pi}{2}$$

$$x = \frac{2\pi n}{5} + \frac{\pi}{10} \checkmark$$

OR $3x = 2\pi n - \left(\frac{\pi}{2} - 2x\right)$

$$x = 2\pi n - \frac{\pi}{2} \checkmark$$

 $(n \in \mathbb{Z})$

(ii) $\cos 3x - \sin 2x = 0$

$$x = \frac{2\pi n}{5} + \frac{\pi}{10} \quad \text{or}$$

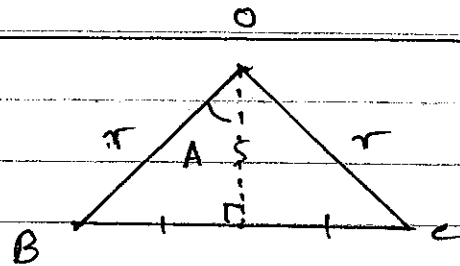
$$x = 2\pi n - \frac{\pi}{2}$$

put $n=0$: $x = \frac{\pi}{10} \checkmark$

is the smallest positive solution.

(b) $\angle BOC = 2A$

(The angle subtended by an arc at the centre is twice the angle subtended by the arc at the circumference.)



$$\sin A = \frac{\frac{1}{2}BC}{r} \checkmark$$

$$\frac{1}{2}BC = r \sin A$$

$$\therefore BC = 2r \sin A.$$

(ii) $|\Delta OBC| = \frac{1}{2}r^2 \sin 2A$

$$|\Delta OCA| = \frac{1}{2}r^2 \sin 2B$$

$$|\Delta OAB| = \frac{1}{2}r^2 \sin 2C$$

$$|\Delta ABC| = \frac{1}{2}r^2 (\sin 2A + \sin 2B + \sin 2C) \quad (1)$$

Also:

$$|\Delta ABC| = \frac{1}{2}BC \times AC \times \sin C$$

$$\begin{cases} BC = 2r \sin A \text{ and} \\ AC = 2r \sin B \text{ from (i)} \end{cases}$$

$$|\Delta ABC| = \frac{1}{2} \times 2r \sin A \times 2r \sin B \times \sin C \checkmark$$

Equate (1) and (2)

$$\frac{1}{2}r^2 (\sin 2A + \sin 2B + \sin 2C) = 2r^2 \sin A \sin B \sin C$$

and

$$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

(as required.)

QUESTION 14----

(c)

(i) $z = \cos \theta + i \sin \theta$

$z - 1 = 1 + \cos \theta + i \sin \theta$

$z + 1 = 1 + (2\cos^2 \frac{\theta}{2} - 1) + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \checkmark$

$z + 1 = 2\cos^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}$

$z + 1 = 2\cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$

(ii) $\frac{z_1 + z_1 z_2}{z_1 + 1}$

$$= \frac{z_1 (1 + z_2)}{(1 + z_1)}$$

using the result from (i) :-

$$= \frac{(\cos \alpha + i \sin \alpha) 2\cos \frac{\beta}{2} (\cos \frac{\beta}{2} + i \sin \frac{\beta}{2}) \checkmark}{2\cos \frac{\alpha}{2} (\cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2})}$$

$$= \frac{\cos \frac{\beta}{2} \operatorname{cis} \left(\alpha + \frac{\beta}{2} - \frac{\alpha}{2} \right) \checkmark}{\cos \frac{\alpha}{2}}$$

$$= \frac{\cos \frac{\beta}{2} \operatorname{cis} \left(\frac{\alpha + \beta}{2} \right)}{\cos \frac{\alpha}{2}}$$

① $\cos \frac{\beta}{2} = 2 \cos \frac{\alpha}{2}$

$\cos \frac{\beta}{2} = 2 \cos \left(\frac{\pi}{2} - \frac{\beta}{2} \right)$

$\cos \frac{\beta}{2} = 2 \sin \frac{\beta}{2}$

$\tan \frac{\beta}{2} = \frac{1}{2} = t$

(iii) $\frac{z_1 + z_1 z_2}{z_1 + 1} = 2i$

$$\frac{\cos \frac{\beta}{2} \operatorname{cis} \left(\frac{\alpha + \beta}{2} \right)}{\cos \frac{\alpha}{2}} = 2i$$

① $\frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}} = 2 \checkmark$

② $\cos \left(\frac{\alpha + \beta}{2} \right) = 0$

$\alpha + \beta = \pi$

$\alpha = \pi - \beta \checkmark$

Sub. into ①

Now $\cos \beta = \frac{1 - t^2}{1 + t^2}$

$$= \frac{1 - 1/4}{1 + 1/4}$$

$$= \frac{3}{5}$$

and $\sin \beta = \frac{2t}{1 + t^2}$

$$= \frac{1}{1 + 1/4}$$

$$= \frac{4}{5}$$

Q14 (c)

10.

$$\therefore z_2 = \frac{3}{5} + \frac{4}{5}i \quad \checkmark$$

$$\textcircled{1} 2 \cos \frac{\alpha}{2} = \cos \beta$$

$$2 \cos \frac{\alpha}{2} = \cos \left(\frac{\pi}{2} - \frac{\alpha}{2} \right)$$

$$2 \cos \frac{\alpha}{2} = \sin \frac{\alpha}{2}$$

$$\tan \frac{\alpha}{2} = 2 = t$$

$$\therefore \sin \alpha = \frac{2t}{1+t^2}$$

$$= \frac{4}{5}$$

$$\cos \alpha = \frac{1-t^2}{1+t^2}$$

$$= \frac{1-4}{5}$$

$$= -\frac{3}{5}$$

$$\therefore z_1 = -\frac{3}{5} + \frac{4}{5}i \quad \checkmark$$